



GameFlow Traffic Control (GTC)

A Game Theory Approach To Traffic Signal Optimization

Group Members

Syed Subhan Shah Bukhari (22P-9447)

Maaz Mustafa (22P-9405)

Moiz Ghazanfar(22P-9425)

Project Supervisor

Mr. Usama Musharaf



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Problem Statement

“Conventional traffic signals use fixed-time cycles, which are inefficient as they cannot adapt to real-time traffic fluctuations, leading to unnecessary congestion and long delays.”



Literature Review: Performance Evaluation

Method	Deployment Status	Evidence
🏆 Game Theory	✅ PROVEN - 350+ Cities	• SCOOT: 350+ cities worldwide (1976-present) • SCATS: Widespread US deployment • Los Angeles: 457 intersections • Blacksburg, VA: 38 intersections
Reinforcement Learning	❌ ZERO Deployments	Chen et al. (2022): <i>"RL-based signal controllers have never been deployed"</i>
Genetic Algorithms	❌ ZERO Deployments	All research confined to simulators <i>"No way to perform feedback in real world"</i>

Literature Review: Performance Comparison

Criterion	Game Theory	RL	GA
Convergence Speed	⚡ 10-50 iterations (seconds)	⌚ Months of training	⌚ 100s-1000s generations
Real-Time Capable	✅ YES	❌ NO	❌ NO
Stability Guarantee	✅ Nash Equilibrium	❌ None	❌ None
Safety	✅ Guaranteed	⚠️ Trial-and-error unsafe	⚠️ No guarantees
Interpretability	✅ Clear payoffs	❌ Black-box	❌ Black-box



The 'Black Box' Problem

Reinforcement Learning:

"Complex and opaque neural networks hard to understand and debug" (Chen et al., 2022)

Cannot explain why a decision was made

Safety-critical systems require transparency

Genetic Algorithms:

"Survival of the fittest" is not an engineering explanation

No clear decision logic

Difficult to validate and tune

Game Theory:

Clear payoff functions: $U = -(\alpha \cdot \text{Wait} + \beta \cdot \text{Delay} + \gamma \cdot \text{Queue})$

Engineers understand WHY decisions are made

Easy to validate and adjust

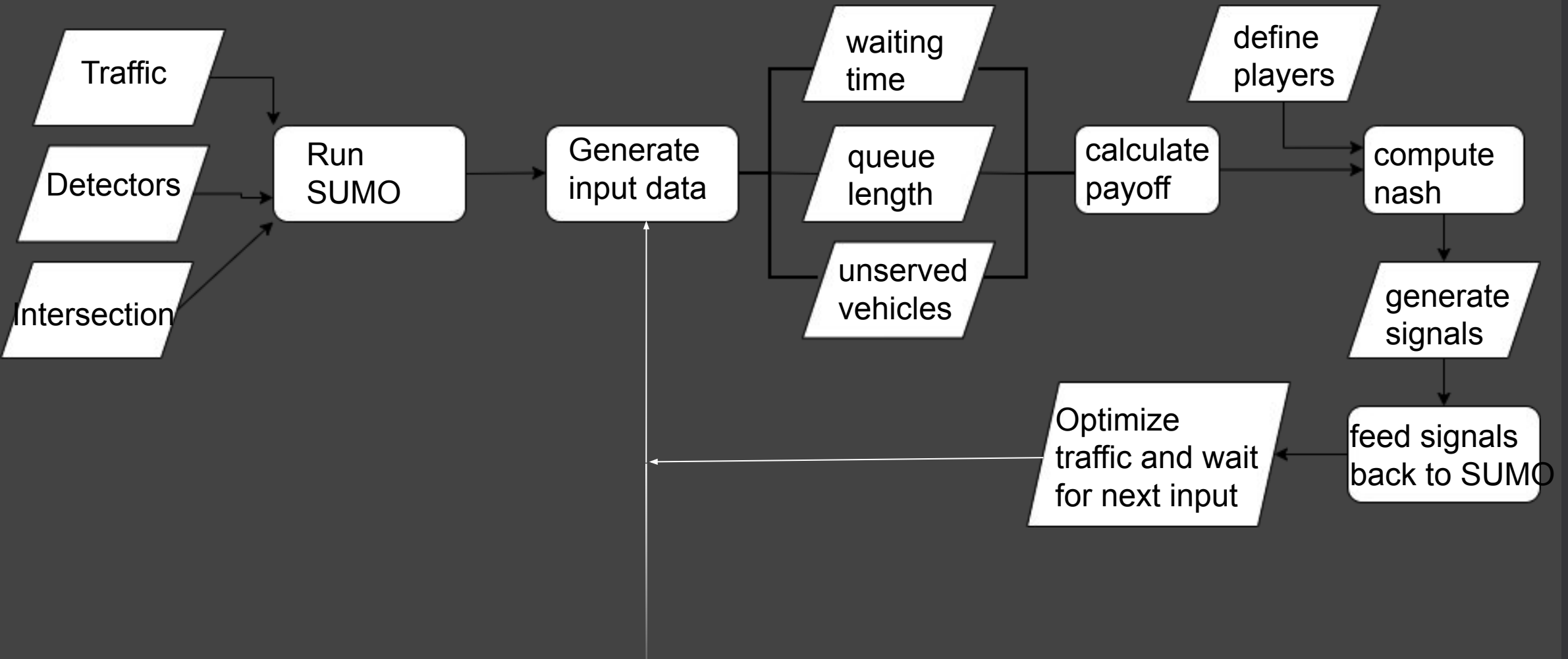
Timeline & Our Contribution

- 1970s ————— SCOOT/SCATS Development (Game Theory-based)
- 1976 ————— First SCOOT Deployment (Glasgow)
- 1980s ————— SCATS Deployment (Australia → US)
- 2000s ————— GA Research Begins (simulation only)
- 2010s ————— RL Research Surge (simulation only)
- 2019 ————— Zaied et al.: 457 intersections (Game Theory)
- 2022 ————— Chen et al. Review: "RL never deployed"
- 2023 ————— Wang et al.: Queue Equilibrium (Game Theory)
- 2025 ————— Our FYP: Game Theory in Pakistan

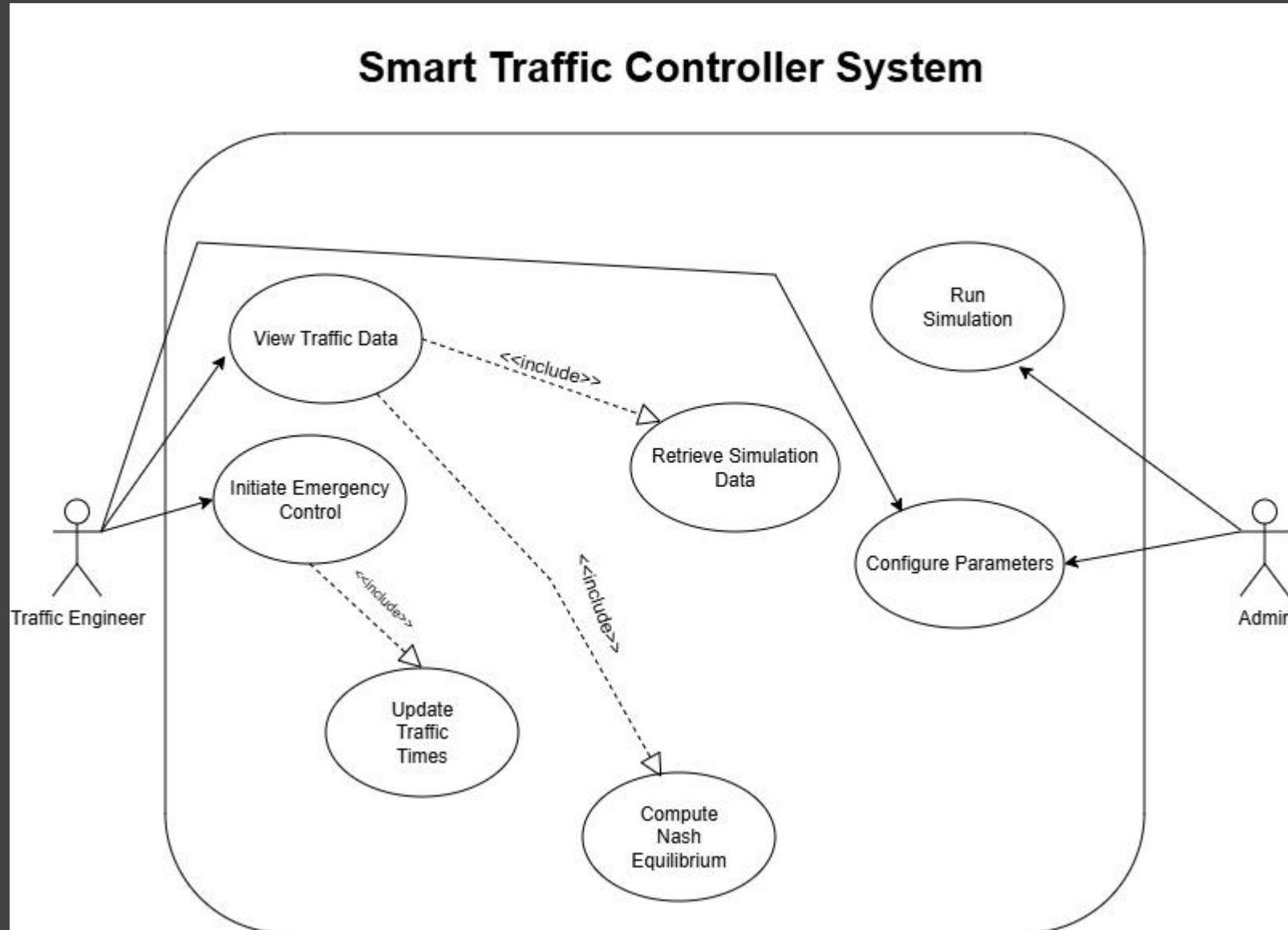
Our Adaptation:

- Modified for multiple timeslots
- Calibrated for mixed traffic
- 2-phase system (NS + EW)
- Nash Equilibrium optimization
- SUMO simulation validation

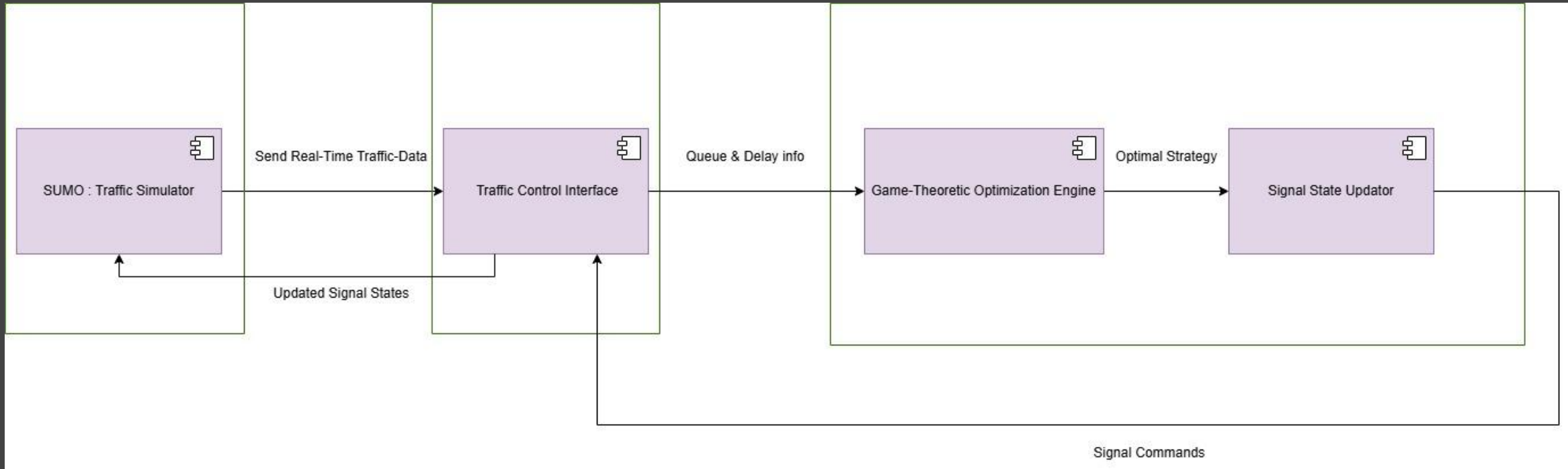
Updated Workflow Diagram



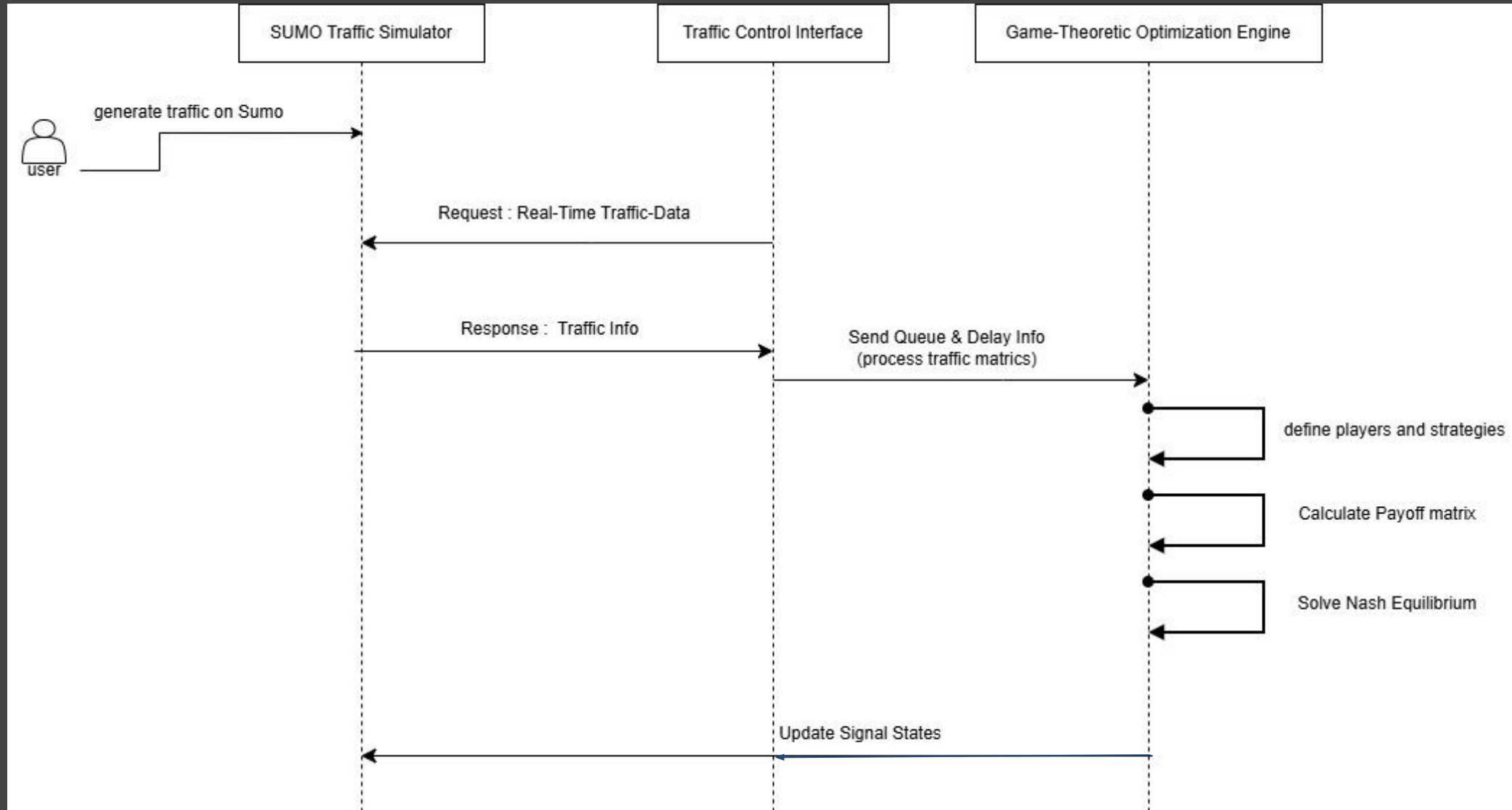
Updated Use-Case Diagram



Architecture Diagram



Activity Diagram





Fixed Timed Signals

(Base Model)



Fixed Timed Model

North & South

Speed: 0.03

Travel Time: 1132.37 seconds.

Teleporting: 36

West & East

Speed: 0.40 to 0.47

Travel Time: ~65–75 seconds.

Spillback along with Unfairness!



Game Theory Solution



Constant Variables:

Type of Traffic: Mix and Aggressive

Time Slots = {Early Morning, Afternoon, Evening}

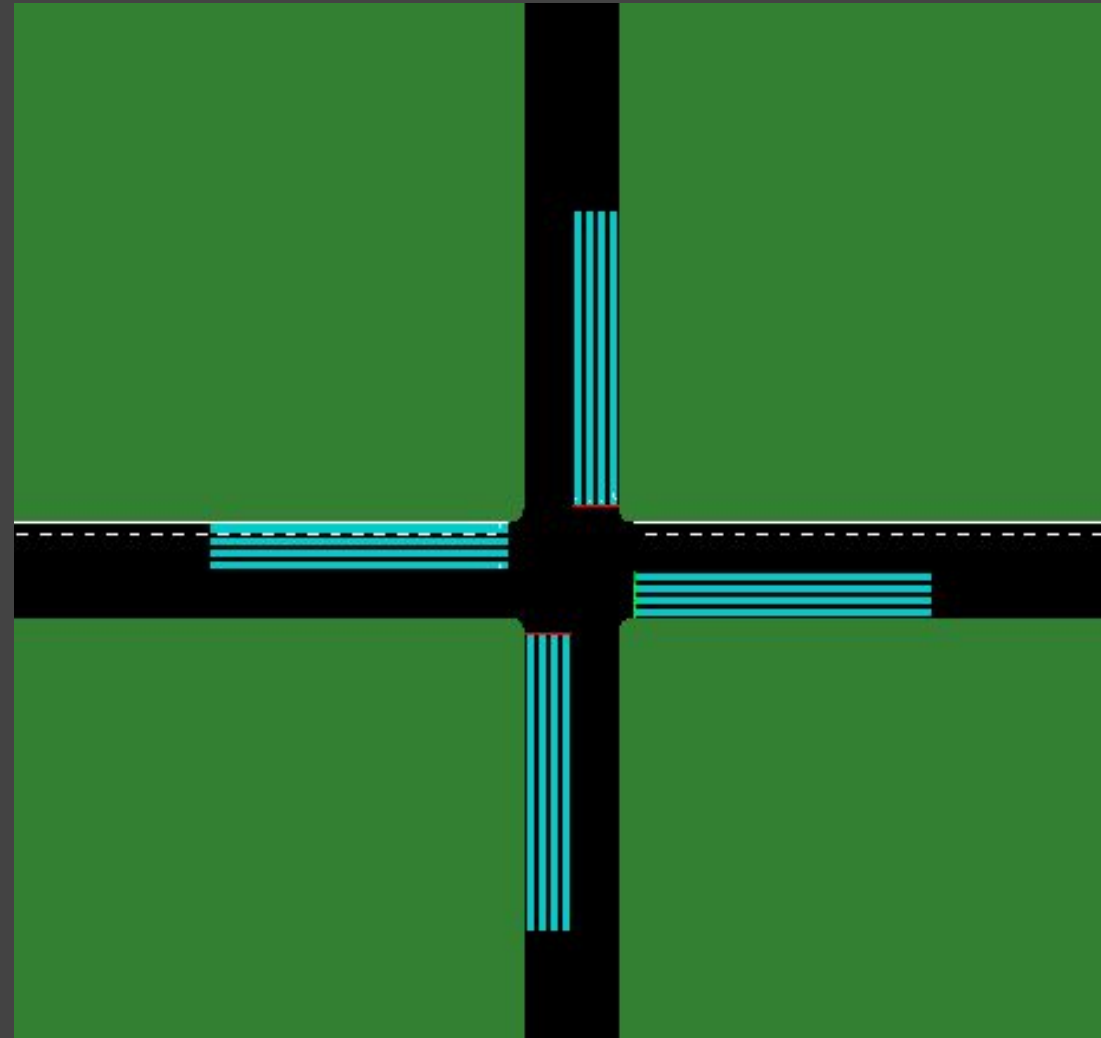
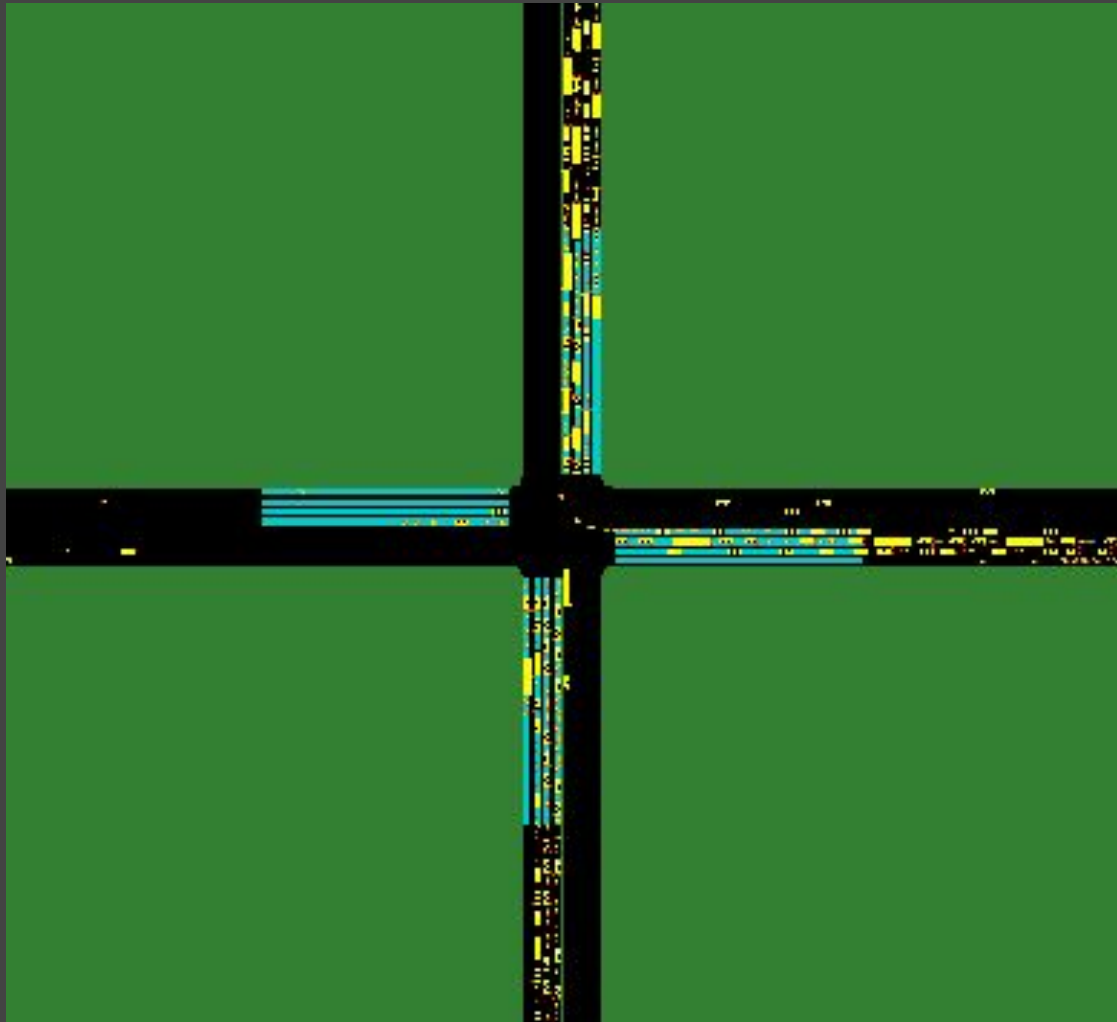
Number of Vehicles per each direction = 10200

Strategy Set = {Red, Green}

Number of player = 2

Payoff = $(\alpha)\text{QueueLength} + (\beta)\text{WaitingTime} +$
 $(\gamma)\text{ResidualVehicles}$

Game Theory vs Base Model



Game Theory at Work!



NS: Demand Green

Crash
 $(-10000.0, -10000.0)$

NS Goes
 $(0, -297.0)$

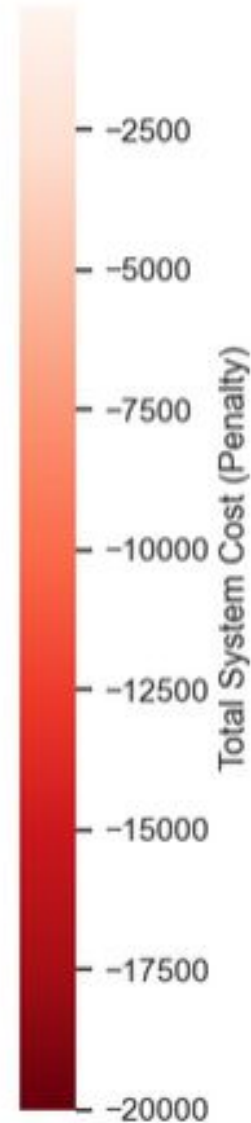
NS: Yield (Red)

EW Goes
 $(-748.5, 0)$

All Red
 $(-748.5, -297.0)$

EW: Demand Green

EW: Yield (Red)



GT vs Base model



```
speed="0.96"  
speed="0.61"  
speed="1.32"  
speed="1.06"
```

```
speed="1.27"  
speed="0.90"  
speed="3.33"  
speed="1.53"
```

GT vs Base model Throughput

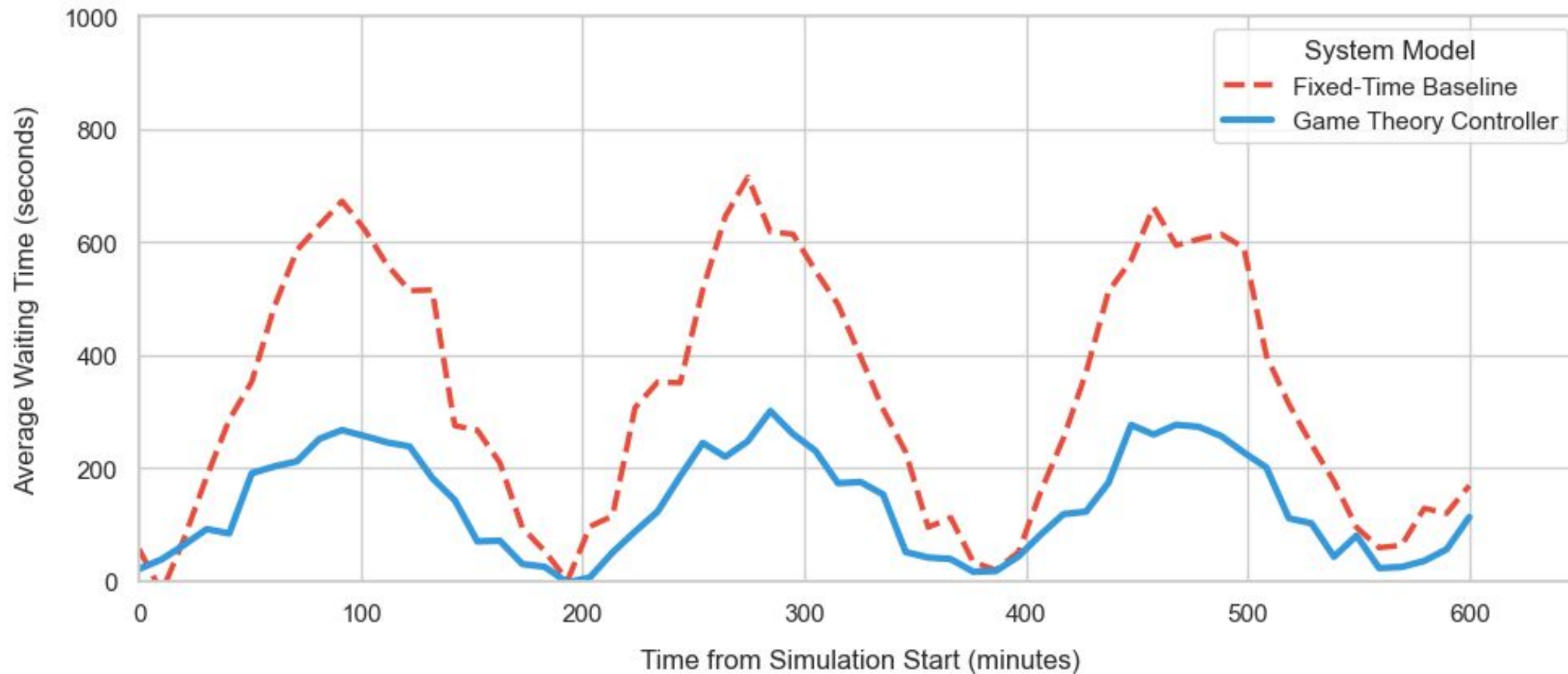


```
departed="8765"  
" departed="9353"  
departed="10048"  
departed="10200"
```

```
departed="10200"  
departed="10200"  
departed="10200"  
departed="10200"
```

Avg Wait Time

**Comparison of Average Waiting Time
(Baseline vs. Game Theory)**





**Game Theory beat
Fixed Timed Signals
by half!!!**

Project App Prototype



Live Metrics



Control Settings

Model Selection

Model Selection: ☒ Real ☐ Simple

Parameters

Parameters:

Parameter	Value
Waiting Time Weight (α)	1
Queue Penalty Weight (β)	0.5
Residual Cost Weight (γ)	0.8
Minimum Green Time	10
Maximum Green Time	60
Green Time Estimate	30

Reset

Project App Prototype



Simulation

Status
● Running

Model
Real

Step
0s

▶ Start

■ Stop

Mode

⏮ Auto

Traffic Control Manager v2.0 | Inspector Edition

Project Planning

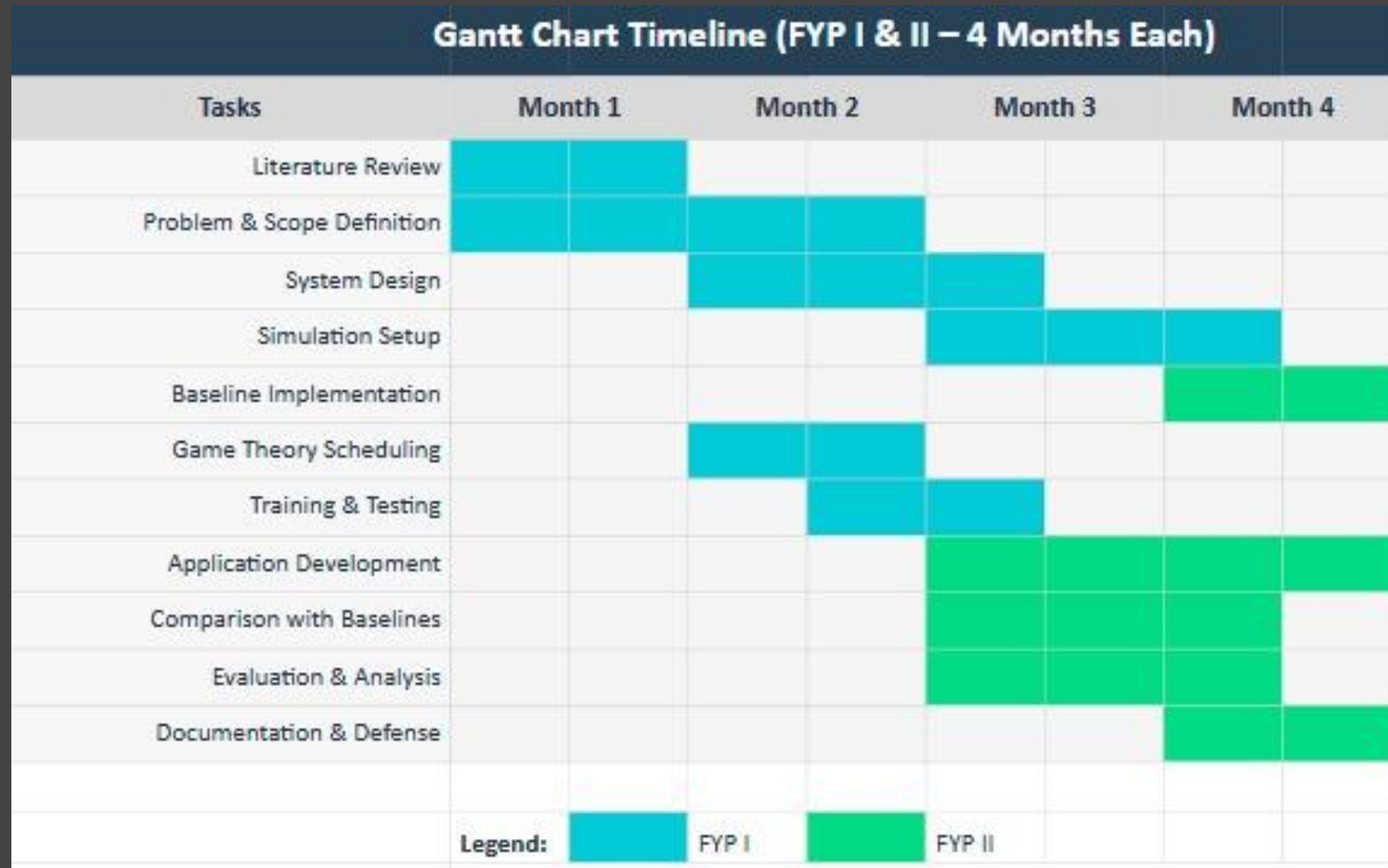


Fig: Gantt Chart

References

Game Theory:

1. **Zaied et al. (2019)** - *Sensors* - 457 intersections, Los Angeles
2. **Wang et al. (2023)** - *Applied Sciences* - Queue equilibrium control
3. **Stevanovic et al. (2009)** - SCOOT/SCATS deployment review
4. **TRL Software (2024)** - SCOOT: 350+ cities worldwide

Reinforcement Learning:

5. **Chen et al. (2022)** - *arXiv* - Comprehensive review
 - **Key Finding:** *"RL-based signal controllers have never been deployed"*
 - Identifies 4 insurmountable deployment barriers

Genetic Algorithms:

6. **Brazilian NSGA2 Study** - *"No way to perform feedback in real world"*
7. Multiple IEEE/ACM papers (2000-2020) - All simulation-based



Thank You!